

Grade Eight Science - Static Electricity Review Questions

MYP Command Terms that are used on this problem set:

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Recall: Remember or recognize from prior learning experience	Construct: Display information in a diagrammatic or logical form
Analyze: To identify parts and relationships, and to interpret information to reach conclusions	Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information

1. **Apply** scientific knowledge to predict the charge on each of the following two items when they are rubbed together (Level 3-4)

Charging situation	Charge on item 1	Charge on item 2
Polyester rubbed with wool		
Glass rod rubbed with silk		

2. **State** the type of charging happening in question 1 above and **state** one way that this type of charging compares to any other type of static charging studied in class (Level 3-4)

[illegible]

3. A plastic rod is rubbed with a piece of silk and then the rod is touched to the ball of a neutral electroscope.

A. **State** the two types of static charging happening here, and **outline** your reasoning below (Level 3-4, Level 5-6)

B. When the plastic rod is rubbed with silk, **state** the charge left on the rod and the silk and **outline** your reasoning (Level 3-4, Level 5-6)

C. A student working in the lab would like to give a neutral electroscope a positive charge. **Apply** scientific knowledge to decide if the student should use the plastic rod or the silk from question "B," above? **Outline** your reasoning below (Level 5-6)

4. **Apply** information to make a scientifically supported judgement to indicate whether unknown "substance X," falls above or below the material given (Level 3-4)

Charging Situation	Is "substance X" below or above the given material on the triboelectric series
Rubbing cotton and "X" together gives "X" a positive charge	
Rubbing wool and "X" together gives "X" a negative charge	

5. A person wraps their hand around a negative styrofoam ball. **Outline** the purpose of doing this and **state** the charges that will be left on the hand and the ball. **Outline** your reasoning using both words and a diagram to support your answer below (Level 3-4, Level 5-6)

Words	Diagram

6. **Recall** that the type of subatomic particle that can move between objects is:

(Level 1-2)

- A. Protons
- B. Electrons and protons
- C. Neutrons
- D. Electrons
- E. Electrons and neutrons
- F. Electrons, protons, and neutrons

7. **Analyse** the following information and answer the question in the box below:

An ebonite rod is rubbed with paper. The paper is then rubbed with a neutral piece of polyester. The polyester is then rubbed with a neutral piece of fur. The fur is then brought near the ball of a neutral electroscope. As the fur is held near, a neutral plastic rod is touched to the aluminum ball of the electroscope. **Apply** your scientific knowledge about static charging to **describe** your reasoning to answer each of the following problems below (Level 7-8)

- A. **Describe** how the ebonite rod, paper, polyester, fur, and electroscope got their charges. Your **description** must clearly indicate the charge left on each object and how each charge was obtained. Scientific language must be used in the **description**.

- B. Using the description in question "A" above, **construct** each of the following diagrams

The induction between the fur and the neutral electroscope with the neutral plastic rod added (Charge transfer must be shown if there is any)	The fur, the plastic rod, and the electroscope with their final charges
	Electroscope:
	Plastic Rod:
	Fur:

8. A neutral ebonite rod is touched to the ball of a positively charged electroscope. Once the rod and electroscope are separated the rod repels a ping pong ball. The ping pong ball then attracts some rice krispies which come near the ping pong ball, but don't touch it.

Apply scientific knowledge to figure out the final charge on the rod, the electroscope, the ping pong ball, and the rice krispies. **Describe** your reasoning below (Level 7-8)

Ebonite Rod	Electroscope	Ping Pong Ball	Rice Krispies

